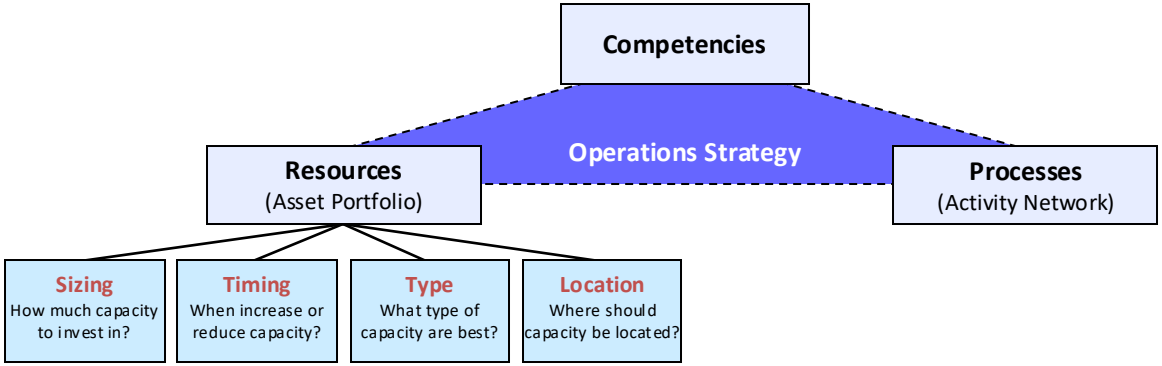
Today's Learning Objectives

- Be able to
  - Explain the concept of capacity strategy and discuss its challenges and importance
  - Describe five different capacity strategies
  - Analyze key capacity drivers
  - Value and optimize capacity
    - Size: marginal analysis
    - Timing: option value of waiting
  - Tailor capacity strategy
    - pick a position in the capacity-inventory-waiting strategy triangle

Capacity Strategy

- A capacity strategy is the long-term plan for *developing resources* that involves decisions on *size, timing, type* and *location* of real assets (refer to class 1 slide 5)



FEDEX TO SPEND $1.8 BILLION TO EXPAND UNIT'S CAPACITY

The FedEx Corporation will spend $1.8 billion to expand its small package carrier, FedEx Ground, by adding 10 hubs and enlarging 23 others. The project will increase the company's capacity by the end of fiscal year 2009, to 4.8 million packages a day from 2.5 million. The first four new hubs will open in Memphis; Dallas; Cincinnati; and Hagerstown, Md., by 2006. Six others will open three years later. More than 300 existing hubs will be expanded or moved. NYT 10/1/2002.

Capacity Strategy: Size and Timing

- Size means deciding capacity investment level
  - All resources are limited by their capacity size, which typically represents the maximal sustainable output rate of that resource
    - Mini_Case_1: SWISS watch making
    - HBS case: ACC and DJC
  - The capacity level at which we choose to operate that resource at any given time decides resource capacity utilization, typically a fraction of the resource capacity size
- Timing means timing of capacity size adjustments
  - There is some leeway about the timing or dynamics of the investment

The trade-offs

- Deciding on capacity size, company incurs various costs when acquiring and operating real assets, and it impacts the whole supply chain
- Key trade-offs are
  - Size: cost of excess capacity vs. cost of capacity shortfall (e.g., opportunity cost)
  - Timing: costs of the capacity adjustment vs. the continuing or expected costs of excess (or shortage)

Example of reversing a capacity investment

Solar-Panel Maker to Close a Factory and Delay Expansion 3 November 2010 The New York Times

Solyndra, a Silicon Valley solar-panel maker that won half a billion dollars in federal aid to build a state-of-the-art robotic factory, plans to announce on Wednesday that it will shut down an older plant and lay off workers.

The cost-cutting move, which will reduce the company's previously announced production capacity, is a sign of the notable shift in the prospects for cutting-edge American solar companies, which now face intense price competition from Chinese manufacturers that use more established photovoltaic technologies.

Just seven weeks ago, Solyndra opened Fab 2, a $733 million factory in Fremont, Calif., to make its high-tech solar panels. The new plant was supposed to be the first phase of a rapid expansion of the company. Instead, Solyndra has decided to shutter the old plant and postpone plans to expand Fab 2, which was built with a $535 million federal loan guarantee. ''Fab 2 is much more efficient and cost-effective than our existing facility,'' Brian Harrison, Solyndra's chief executive, said in an interview. ''We're adjusting our plans to be more in line with where the market is and where our business is at the moment.''

When Solyndra filed for an initial public stock offering in December, it estimated it would have a total production capacity of 610 megawatts by 2013 if its two plants were fully built out. The company now expects it have capacity of 285 to 300 megawatts by 2013. The company is the most prominent of a wave of Silicon Valley solar start-ups that hoped to transform the economics of the industry. Gov. Arnold Schwarzenegger of California and Energy Secretary Steven Chu helped break ground on Fab 2 last year, and President Obama made an appearance at the unfinished factory in May to extol Solyndra's innovative technology.

Mr. Harrison noted that the market had undergone a significant shift since Solyndra filed for the stock offering, with solar module prices plummeting as low-cost Chinese manufacturers like Suntech and Yingli ramped up production. That has put pressure on companies like Solyndra, which makes advanced thin-film solar modules that are less efficient than conventional photovoltaic modules but had been cheaper to install until prices began to fall sharply last year. Solyndra said it would lay off around 40 employees and not renew contracts for about 150 temporary workers as a result of the consolidation. The closing of the old factory, called Fab 1, will save the company more than $60 million in capital expenditures, executives said.

Mr. Harrison, who became Solyndra's chief executive in July, said that despite the cutbacks, the company's production of solar panels for commercial rooftops would double in 2011 from the previous year. He said Solyndra continued to receive large orders from customers. Depending on how the market evolves, Solyndra could reopen Fab 1 or expand its new factory, Mr. Harrison said.

Example of reversing a capacity investment

Eli Lilly cancels manufacturing project in Prince William

Pharmaceutical giant Eli Lilly has abandoned a $325 million development in Prince William County, VA, citing lack of demand for its insulin products. The company's project on a 120-acre site at the Innovation @ Prince William Technology Business Park was considered the centerpiece of the county's effort to become a prominent bioscience location. Eli Lilly has been building a 300,000-square-foot insulin manufacturing facility that was scheduled to be completed in 2009 and bring 350 jobs to the county. Indianapolis-based Eli Lilly has about 120 workers in Prince William who will be offered opportunities at other sites or be given a severance package if they choose not to say with the company.

The decision to shut down the project is a "difficult announcement" and "part of a broader restructuring of our manufacturing operations," says company spokesman Ed Sagebiel. The current demand for Eli Lilly's insulin products can be met at existing sites and a facility being built in Italy, company officials say. Construction at the Prince William site will stop immediately.

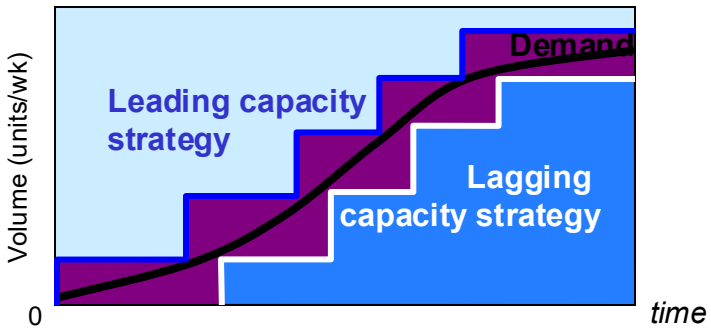
Eli Lilly chose Prince William County from a list of nearly 300 sites when it announced plans to build the facility in May 2002. At the time, the company planned a $425 million, 600,000-square-foot facility with 700 employees, but in 2005 Eli Lilly cut the project's size in half because of reduced demand for its products.

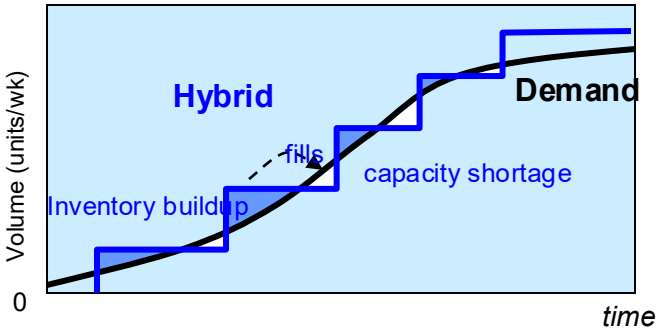
The company (NYSE: LLY) received more than $4 million in local and state incentives to purchase and develop the property, and those funds must be returned.

Washington Business Journal - January 11, 2007 **by Joe Coombs and Neil Adler Staff Reporters** http://washington.bizjournals.com/washington/stories/2007/01/08/daily41.html

Five capacity (size & timing) strategies

1. Demand Chasing
2. Leading
3. Lagging
4. Hybrid timing
5. Follow-your competitor





<u>Decisions to make:</u>

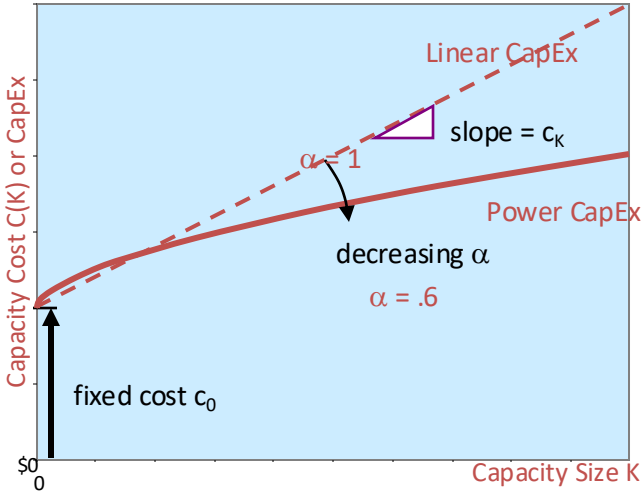
1. Size = by how much? (many small sizes vs. one big size, lumpiness)
2. Time = when? (lead or lag)

Ways to Expand capacity

- Building a new facility (Greenfield)
- Expanding a current facility (Brownfield)
- External capacity provision
  - Outsourcing
  - Subcontracting
  - Franchising
  - Alliances
- Process improvement and restructuring at current facility
- Adding a shift or working days
- Tactical capacity adjustment
  - Aggregate planning: seasonal demand fluctuations can be buffered by
    - working overtime or adding temporary workers;
    - by building inventory or keeping a waiting list for back orders

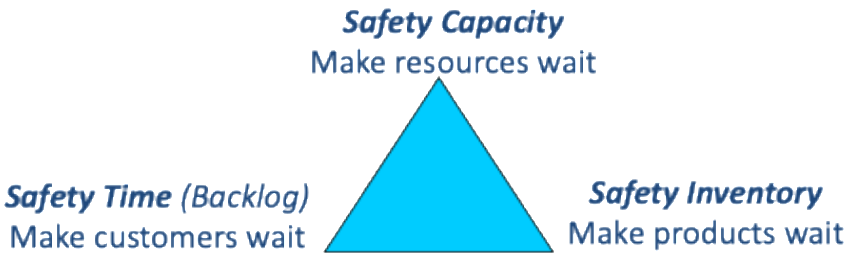
Key Capacity Drivers

- Economies of Scale (EOS)
  - Bigger is Better: e.g., cost of doubling the size is less than double the cost of the original size, i.e., $C(2K) < 2C(K)$ where
    - K is size of Capacity
    - C(K) is the associated capacity investment cost
  - Two useful capacity investment cost models
    1. Linear CapEx: $C(K) = c_0 + c_K K$ where
       - c_0 is fixed capacity investment costs
       - c_k is marginal capacity investment costs
    2. Power CapEx: $C(K) = c_0 + (c_K/\alpha)K^{\alpha}$
       - $0<\alpha<1$, allowing for decreasing marginal cost
       - c_K is marginal capacity investment costs at K=1



Key Capacity Drivers (Cont.)

- Uncertainty
  - uncertainty degrades the value of capacity: capacity size constraint prevents company from enjoying upside demand opportunity while it is fully exposed to the downside risks
  - To counteract its impact, we must either invest in safety capacity or make either customers or products wait (the triangle below)
- Capacity frictions such as lead-time, lumpiness & fixed costs
  - Long LT make it not possible for a capacity leading strategy
- Discount factor, r
- Irreversibility
- Competitors
- Technology



Value and Optimize Capacity Size

- Under the presence of uncertainty, the key trade-off is the cost of excess capacity vs. cost of capacity shortfall
  - excess capacity costs is marginal capacity investment costs c_k
  - capacity shortfall costs includes
    - the unit operating profit margin, lets use m for notation
    - the unit opportunity loss, let's use c_p for notation
- To decide the optimal capacity size, we may use marginal analysis:
  - Marginal capacity costs is c_k
  - Marginal capacity value is $(m + c_p)\text{Prob}(\text{Demand}>K)$
- So, optimal K should be at where $(m + c_p)\text{Prob}(\text{Demand} > K) = c_k$
    - Which is equivalent to $\text{Prob.}(\text{Demand} \le K) = \frac{m+c_p-c_k}{m+c_p} = F(K)$, implies
    - Optimal $K = F^{-1}(\frac{m+c_p-c_k}{m+c_p})$ where F(K) is the cumulative demand distribution

Value and Optimize Capacity Size

- Relevant costs and demand distribution leads to Optimal Capacity size K:
  1. If Demand is Discrete, use the cumulative demand distribution table, i.e., F(D), so that cumulative probability at K is more than or equal to $\frac{m+c_p-c_k}{m+c_p}$
  2. If Demand is N(μ, σ^2)
     - K = μ+z*σ where z = Norm.S.Inv($\frac{m+c_p-c_k}{m+c_p}$) or
     - K = Norminv($\frac{m+c_p-c_k}{m+c_p}$, μ, σ)
- Insights and relate to the triangle:
  - If marginal capacity cost is very small, i.e., $\frac{m+c_p-c_k}{m+c_p}$ > 50%, the optimal capacity size should be more than the average demand with safety capacity (e.g., z*σ for Normal demand distribution case); safety capacity can be interpreted as making resources wait
  - If the marginal capacity cost is prohibitively expensive, i.e., $\frac{m+c_p-c_k}{m+c_p}$ < 50%, the optimal capacity size should be scaled down (below average demand) which can be interpreted as making customers wait,
  - If the marginal capacity cost and value-add are equal, i.e., $\frac{m+c_p-c_k}{m+c_p}$ = 50%, then it's best to size capacity at the median demand (50th percentile) level

Challenges for Capacity Size

- Capacity size is a soft, malleable constraint
  - It is often a less precise, soft constraint, the point beyond which marginal production costs increases sharply
- Capacity is like “black art”; it depends on OM tactics
  - It depends on all operational polices used in activity network (e.g., aggregate planning), actual capacity utilization depends on everything…
  - Utilization =FR/capacity, Flow Rate is the real production output
- Measuring and valuing capacity shortfall is not obvious
  - Customer wait, Substitution, Lost sale, Loss of goodwill and future demand
- Capacity decisions can be political
  - Typically, a firm invests in multiple types of resources that have different financial and operational properties
  - They affect many stakeholders
- Capacity investment involves long-run planning under Uncertainty
  - There is uncertainty over the future rewards from the investment
  - Arguably this is the greatest challenge for capacity strategy

Value and Optimize Capacity Timing

- Strategic timing
  - often delas with the essential question: should we adjust the capacity size now or should we wait?
  - captures key capacity investment drivers (EOS and uncertainty) over longer time frames
  - often suppresses tactics such as inventory carry-over.
  - Tool to quantify the financial benefits of waiting
    - Option value of waiting: NPV(wait)-NPV(do not wait)
- Tactical timing
  - captures key capacity adjustment tactics
    - When labor is reasonably “frictionless” and adjustable, changing the number of workers, working hours, shifts, working days is an effective and quick lever to adjust capacity level
    - Advance inventory build up or a waiting list (back-orders) can also accommodate temporary demand surges
  - Deciding on tactical capacity adjustments is known as
    - Aggregate planning

Strategic Timing: a stylized example

The Value of Capacity Expansion Postponement

Acer, a Taiwanese computer company, started operations by serving the lower-price segments in Asia. Later, in its quest for a global expansion strategy, it ruled out competing in the high-price segments served by established PC giants. Rather, it decided to expand into emerging markets by serving other low-price segments in Mexico, South-Africa, and Russia.

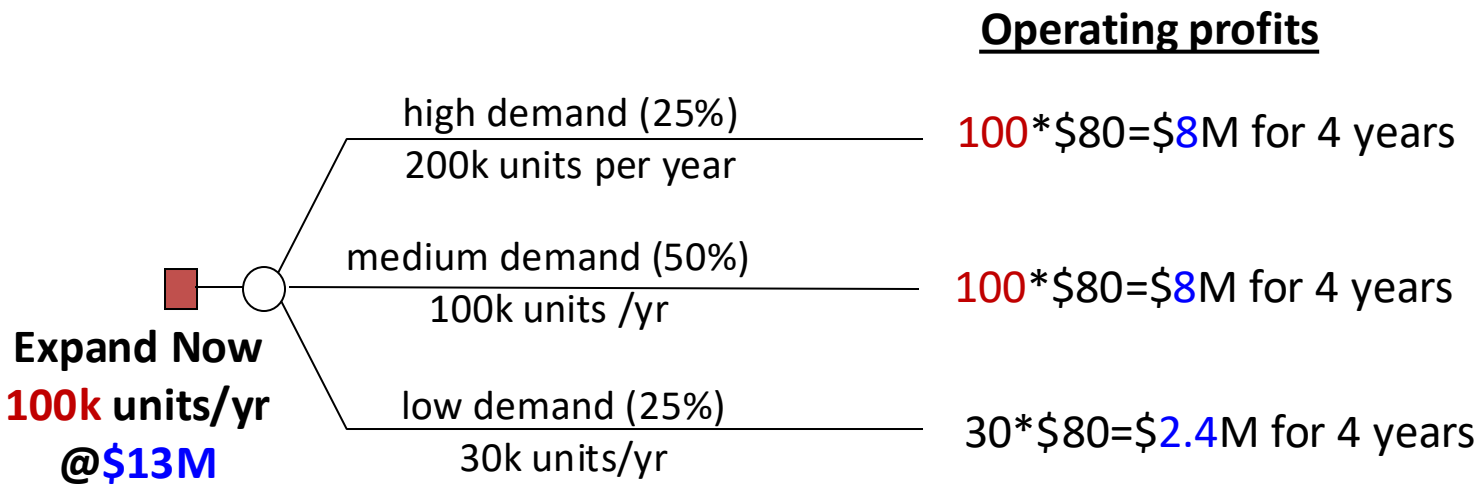
Postponing its global expansion had two benefits:

1. It allowed Acer to reduce its initial capital requirements.
2. The company could learn more about the product and service needs of its Asian customers. With this knowledge, it could size and tailor its later capacity expansion to better fit the needs of these emerging markets.

Strategic Timing: a stylized example

- Demand from these emerging markets is highly uncertain; marketing and sales reports predict that Acer's low-priced PC will either be a blockbuster, a success, or a dud.
  - Assume that the demand forecast for these three scenarios is, respectively: 200 thousand units per year with likelihood of 25%, 100 thousand units per year with 50% likelihood, or 30 thousand units per year with 25% likelihood.
- The cost structure is assumed to be as following:
  - capacity expansion incurs a fixed cost of $8 million plus a marginal cost of $50 per unit; e.g., adding production capacity of 100,000 units would costs 13 million (=8,000,000+50*100,000).
  - Profits contribution is $80/PC
  - Capacity shortfall costs $20/PC
  - Discount rate is 25%
  - The process and product technology is commercially viable for four years (at that point a new technology would be needed, an issue we will ignore for now).
- Should Acer wait one year?

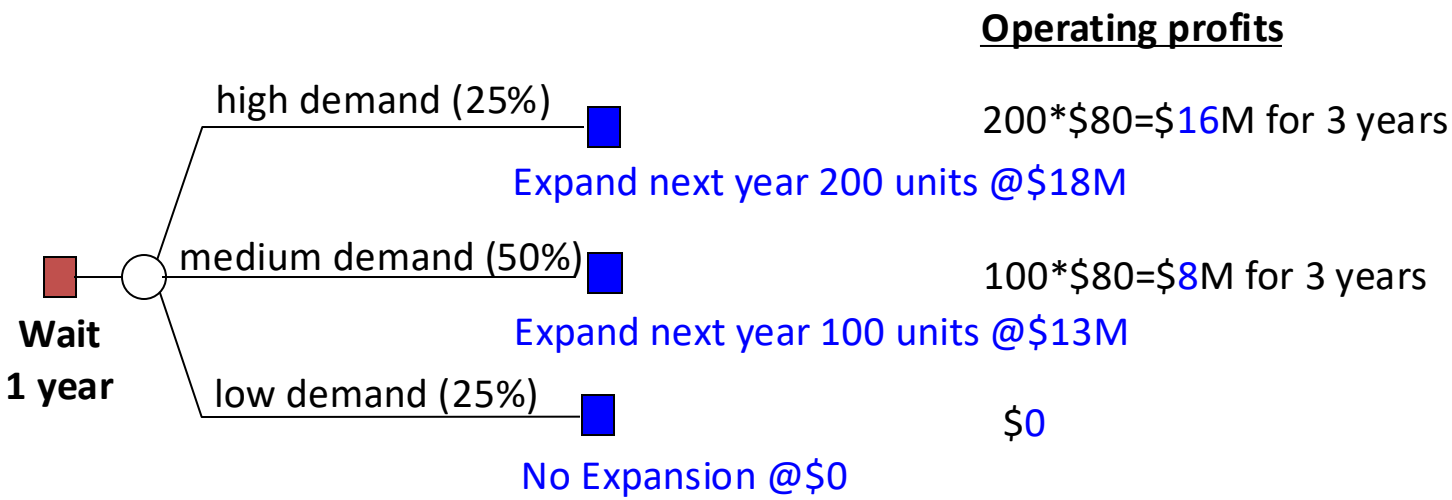
Strategic Timing: a stylized example



- Expected value (expand now, no wait) per year
  - = 25%*8 + 50%*8 + 25%*2.4 = $6.6M/yr.
- with discount rate 25%, NPV(Expand now, no wait)
  - = -13 + 6.6/1.25 + 6.6/1.25^2 + 6.6/1.25^3 + 6.6/1.25^4

    = $2.59M

Strategic Timing: a stylized example



- Expected value (wait) per year
  - = 25%*16 + 50%*8 + 25%*0 = $8M/yr.
- with discount rate 25%, NPV(wait)
  - = -11/1.25 + 8/1.25^2 + 8/1.25^3 + 8/1.25^4 = $3.69M
    - Expected capital expenditure of wait for one year is 25%*18 + 50%*13 + 25%*0 = $11M

Strategic Timing: a stylized example

- In summary:
  - NPV(no wait) = $2.59M
  - NPV(wait) = $3.69M
- Option value of waiting = NPV(wait) - NPV(no wait)

  = 3.69M - 2.59M = $1.1M
- Recommendation: wait!

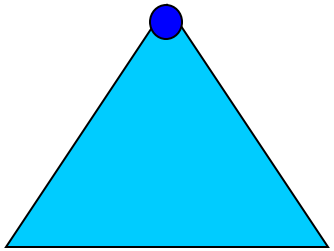
Strategic Timing: The Option Value of Waiting

- The value of waiting lies in more accurate forecasts
  - During the waiting period companies may perform several tests, or observe what happens elsewhere, to achieve better forecasts
- Waiting is valuable if
  - Uncertainty can be reduced substantially
  - Investment opportunity does not disappear , i.e. there is a long enough product life cycle
- Waiting is problematic if
  - First mover advantage is important
  - Production learning (cost reduction through experience) is important

Tailor capacity strategy

- The Capacity-Inventory-Waiting Triangle
  - If the cost of capacity is small compared to the cost of make customers waiting, excess capacity will be put in place, otherwise customers will be asked to wait.
  - In seasonal industries, if the cost of capacity is small compared to the cost of holding inventory, capacity can be sized to meet peak demand (making resources wait), o/w capacity is sized smaller than peak demand and inventory is built (make products wait) during the low season to meet peak demand.
  - ……

Safety Capacity
Make resources wait



Safety Time *(Backlog)*
Make customers wait

Safety Inventory
Make products wait

Summary of Learning Objectives

- Explain the concept of capacity strategy and discuss its challenges and importance
  - Capacity strategy: long-term strategy for investing in resources (size, timing, type, location)
    - Key trade-off in sizing is cost of excess capacity vs. the cost of capacity shortfalls
    - Key trade-off in Timing is cost of the capacity adjustment vs. the continuing or expected costs of excess (or shortage)
- Discuss the five different capacity strategies and their implementations
  - Demand chasing is hard to implement because of frictions (lead-times, lumpiness, fixed costs)
  - Leading strategies offer high service levels. They are a good fit for companies that offer innovation and service.
  - Lagging strategies offer high efficiency and are less risky. They are a good fit for companies that focus on low prices.
  - A hybrid between leading and lagging uses inventory to improve service and efficiency.
  - A fifth strategy is to strictly follow a competitor's strategy
- Analyze key capacity drivers
  - Economies of scale lead to larger capacity adjustments but pushes for delayed expansion timing
  - Uncertainty decreases the value of capacity investments but pushes for larger investment increments
  - Reduced capacity LT, lumpiness and irreversibility lead to higher NPV
  - High discount factors lead to smaller capacity adjustments
- Value and optimize capacity strategy
  - Optimal safety capacities for resources are such that the likelihood that an asset is a bottleneck is equal to the ratio of its marginal capacity cost to its marginal value, i.e., Prob(Demand > K) = $\frac{c_k}{m+c_p}$
  - When demand is highly uncertain it may pay off to gather more information first before making the investment. Strategic waiting (and information gathering) can be part of a low-cost, lagging strategy
  - Aggregate planning can be used for tactical capacity adjustments (using a spreadsheet)
- Tailor capacity strategy by picking a position in the capacity-inventory-waiting triangle
  - A good capacity size strategy uses a tailored combination of safety capacity, safety inventory, and safety time to mitigate value degradation due to volatility.
  - Making products wait (seasonal inventory) or making customers wait are substitutes for investing in safety capacity; a combination of the three strategies can mitigate the effects of volatility